

LatentArena: AI-driven Content Prediction Platform

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Abstract

LatentArena introduces a novel decentralized platform combining content creation, state-of-the-art AI agents, and prediction markets. The platform implements a three-party system consisting of content creators, multimodal AI judges with distinct personalities, and users making range-based predictions. This document presents the technical architecture, game-theoretical foundations, and economic models underlying the platform. We specify the platform's Game theory properties and economic sustainability under various market conditions.

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1 Introduction

1.1 Problem Statement

The current landscape of content and social media platforms operates on a model where user attention is commoditized for targeted advertising. Content consumers, despite being the cornerstone of platform engagement, receive no direct monetary compensation for their attention and participation. Content creators face a bifurcated reality - they must reach a critical mass of followers before monetization becomes viable, creating an inherently inequitable system where rewards are concentrated among a small percentage of top creators. This model not only undermines fair value distribution but also fails to incentivize quality content creation and meaningful engagement from day one.

1.2 Solution Overview

Latentarena reimagines the content platform paradigm by introducing a prediction market mechanism that aligns creators and consumers without advertisements. Unlike traditional platforms that rely on social reinforcement mechanisms (likes, shares, comments) to drive engagement, Latentarena utilizes prediction markets to maintain user interest. The platform addresses the absence of prediction markets in content evaluation by enabling strategic forecasting of content performance.

The platform employs AI agents as impartial judges with distinct evaluation frameworks. Content creators monetize their work through self-assessment and stake-based mechanisms, removing traditional monetization barriers. Users function as active stakeholders through range-based predictions, converting platform engagement into potential returns. This integration of content creation, AI evaluation, and prediction markets establishes a system where content quality emerges through market forces, with earnings distributed based on contribution accuracy.

1.3 System Architecture

1.3.1 Content Creators

Content creators serve as the primary value generators in the ecosystem. They upload video content and participate in the prediction mechanism through self-assessment. Notably, creators can earn from their first upload through a combination of accurate self-assessment and community engagement, breaking the traditional barrier of requiring large follower bases for

monetization.

1.3.2 AI Judges

A panel of 8 AI judges performs content assessment. These judges operate independently, providing quantitative scores and qualitative feedback. The system regularly rotates judges and introduces new personalities to maintain evaluation diversity and prevent gaming of the system. For each content evaluation, four judges are randomly selected from the pool.

1.3.3 Users

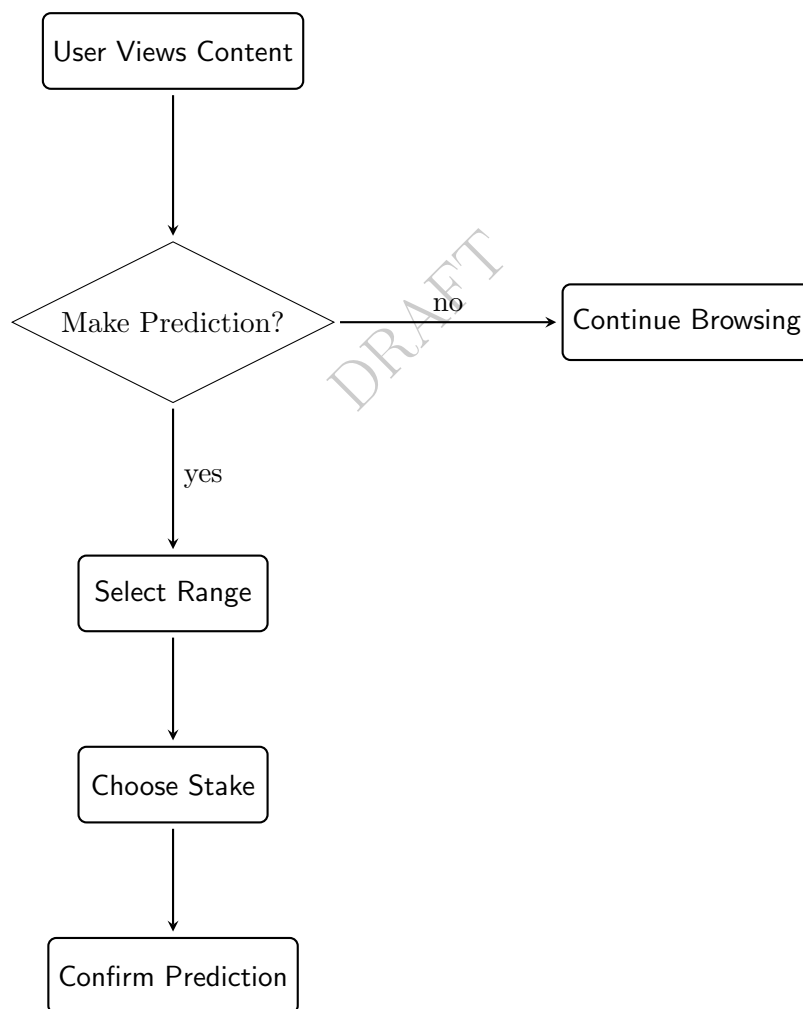


Figure 1: User Prediction Flow

Platform users participate through the prediction mechanism. They can:

- Select from two stake tiers
- Choose from multiple prediction ranges (0.1, 0.25, 0.5, 1.0)
- Access judge commentary
- Earn rewards based on prediction accuracy

2 Platform Mechanics

2.1 Content Flow

The content lifecycle on Latentarena follows a structured path designed to ensure quality, fairness, and engagement:

1. Submission Phase

- Creator uploads video content (2 minutes)
- Content undergoes automated format verification
- Initial guideline compliance check
- Optional self-scoring by creator

2. Verification Phase

- Manual review for guideline adherence
- Content fingerprinting for originality
- Technical quality assessment
- Creator verification

3. Prediction Window

- 24-hour open window for predictions
- Range-based prediction submissions
- Minimum participant threshold monitoring

4. Evaluation Phase

- Random selection of 4 AI judges
- Independent score submission
- Commentary generation
- Score aggregation and rounding

5. Settlement Phase

- Final score calculation
- Reward distribution
- Content archival
- Analytics generation

2.2 Prediction Mechanism

The platform implements a range-based prediction system with corresponding multipliers:

Range	Multiplier	Risk Level
0.1	8x	Highest
0.25	4x	High
0.5	2x	Medium
1.0	1x	Low

Table 1: Range-Multiplier-Risk Relationship

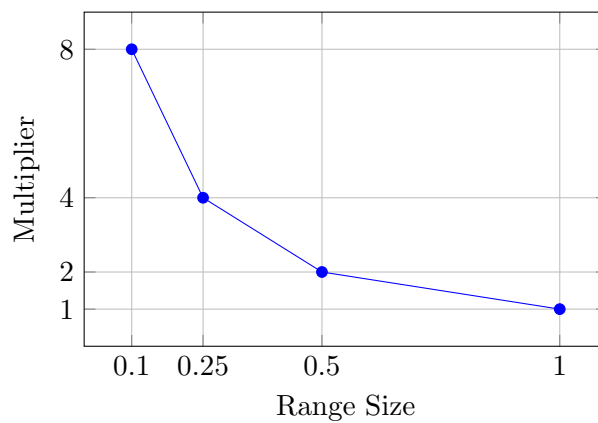


Figure 2: Range-Multiplier Relationship

3 Mathematical Framework

3.1 Score Distribution Model

The platform employs a Beta distribution to model the score distribution, providing a more realistic representation of judge scoring patterns than a uniform distribution. For a score $s \in [0, 10]$:

$$X \sim \text{Beta}(\alpha, \beta) \quad (1)$$

$$f(s) = \frac{s^{\alpha-1}(1-s)^{\beta-1}}{B(\alpha, \beta)} \quad (2)$$

where $B(\alpha, \beta)$ is the Beta function and parameters α, β are estimated from historical scoring data.

3.2 Reward Distribution Model

The reward system incorporates both range overlap and multiple winner scenarios. Let:

- S = total stake pool
- R_i = selected range for participant i
- $M(R_i)$ = multiplier function for range R_i
- $N(R_i)$ = expected number of winners in range R_i

The multiplier function remains:

$$M(R_i) = \begin{cases} 8, & R_i = 0.1 \\ 4, & R_i = 0.25 \\ 2, & R_i = 0.5 \\ 1, & R_i = 1.0 \end{cases} \quad (3)$$

Expected value calculation accounting for overlaps:

$$EV(R_i) = P(\text{win}|R_i) \times M(R_i) \times S \times \frac{1}{N(R_i)} \quad (4)$$

where:

$$N(R_i) = n \times P(R_i \cap R_j \neq \emptyset) \quad (5)$$

$$P(R_i \cap R_j \neq \emptyset) = \int_{R_i} f(s)ds \quad (6)$$

4 Game Theory Analysis

4.1 Sequential Equilibrium Framework

The platform implements a dynamic game with incomplete information. For each prediction round:

- Information sets h_i for each player i
- Belief system $\mu(h)$ at each information set
- Strategy profile $\sigma = (\sigma_1, \dots, \sigma_n)$

Sequential rationality condition:

$$EV_i(\sigma_i|h, \mu) \geq EV_i(\sigma'_i|h, \mu) \quad \forall \sigma'_i, h \quad (7)$$

Belief consistency requirement:

$$\mu(h) = \lim_{k \rightarrow \infty} P(h|\sigma^k) \quad (8)$$

where $\{\sigma^k\}$ represents a sequence of totally mixed strategies.

4.2 Signaling Game Analysis

The creator self-scoring mechanism forms a signaling game:

- Creator signal: $s_c \in [0, 10]$
- True quality: $\theta \in [0, 10]$
- Signal cost: $c(s_c, \theta)$

Separating equilibrium conditions:

$$\frac{\partial c}{\partial s_c} > 0 \quad (\text{costly signaling}) \quad (9)$$

$$\frac{\partial^2 c}{\partial s_c \partial \theta} < 0 \quad (\text{single crossing}) \quad (10)$$

4.3 Dynamic Pool Analysis

The stake pool distribution evolves according to:

$$\Delta S_i(t) = \alpha[EV_i(t) - \overline{EV}(t)]S_i(t) \quad (11)$$

where:

- $S_i(t)$ = stake in range i at time t
- α = adjustment rate
- $\overline{EV}(t)$ = average expected value across ranges

4.4 Economic Equilibrium

Market depth requirement:

$$S_{min} = \max(2T_{min}, \kappa\sigma_X) \quad (12)$$

where:

- T_{min} = minimum stake tier
- κ = market depth parameter
- σ_X = score volatility

Platform revenue condition:

$$E(R) = 0.025 \sum_{i=1}^n S_i + \sum_{i=1}^n P(no_winners)_i \times R_{p_i} \geq C_{op} \quad (13)$$

where C_{op} represents operational costs.

5 Technical Implementation

5.1 Content Management

5.1.1 Video Specifications

- Maximum duration: 120 seconds
- Supported formats: MP4, MOV, AVI

- Maximum file size: 100MB
- Minimum resolution: 480p
- Frame rate: $\geq 24\text{fps}$
- Audio codecs: AAC, MP3

5.1.2 Storage Architecture

- Primary storage: Cloud-based distributed system
- Content addressing: SHA-256 hashing
- Future integration: IPFS implementation
- Redundancy: 3x replication

5.2 AI Judge System

5.2.1 Personality Framework

Each judge maintains:

- Distinct evaluation criteria
- Consistent personality traits
- Specialized domain expertise
- Natural language generation capabilities

5.2.2 Scoring Protocol

- Individual scores rounded to 0.1
- Automated aggregation and averaging
- Personality-driven commentary

Type	Focus Areas	Evaluation Criteria
Technical	Quality, Composition	Video stability, lighting, audio clarity
Creative	Innovation, Style	Originality, artistic elements, storytelling
Engagement	User Interest	Viewer retention potential, call-to-action effectiveness
Content	Subject Matter	Topic relevance, information accuracy

Table 2: AI Judge Evaluation Framework

5.3 Content Piracy

LatentArena maintains a strict policy against content piracy. If submitted content is found to be pirated:

- Creator forfeits all earnings and staked tokens
- Platform permanently bans the creator’s address
- User predictions and earnings remain unaffected
- Forfeited creator stakes are added to the community treasury

6 Conclusion

LatentArena represents a paradigm shift in content valuation and engagement, combining state-of-the-art AI evaluation with decentralized prediction markets. The platform’s mathematical foundations ensure fair participation and sustainable operation, while its technical architecture supports scalable growth and evolution.

7 Technical Specifications

7.1 System Parameters

7.1.1 Content Specifications

- Video duration: 120 seconds
- Maximum file size: 100MB

- Minimum resolution: 480p
- Frame rate: $\geq 24\text{fps}$
- Supported codecs: H.264, H.265
- Audio formats: AAC, MP3
- Content hash: SHA-256

7.1.2 Temporal Parameters

- Prediction window: 24 hours
- Judge scoring period: 1 hour
- Settlement period: 2 hours
- Content review: ≤ 12 hours
- Judge rotation: 168 hours (weekly)
- Minimum time between submissions: 1 hour

7.1.3 Economic Parameters

- Platform fee: 2.5%
- Minimum participants: 2
- Tier 1 stake: T1 TOKENS
- Tier 2 stake: T2 TOKENS
- Judge interaction fee: T_i TOKENS

7.2 Prediction Parameters

- Range options: [0.1, 0.25, 0.5, 1.0]
- Multipliers: [8x, 4x, 2x, 1x]
- Creator fixed range: 0.5
- Score precision: 0.1

7.3 Judge System Parameters

- Minimum judge pool size: 8
- Active judges per content: 4
- Personality traits per judge: ≥ 5
- Response time limit: 1 hour
- Commentary length: 50–200 words
- Score aggregation method: Arithmetic mean

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